bbx ANALYZER

Flight report · v2.4 · logs turned into answers

Fawkes

BTFL_FAWKES_20260927_192610.BBL

274.2 s · 1010 Hz · Betaflight 2025.12.0-RC4

PROPS	ALL-UP WEIGHT	HOVER	HOVER POWER	THRUST / WEIGHT
7″ × 2	**536 g**	**109 Hz**	**≈54 W**	**8.6×**
set by you	estimated	23.8% motor command	momentum theory	extrapolated

Key numbers

Noise left after the gyro filters	**1.9** °/s RMS
D-term noise	**6.11**
Noise reaching the motors	**1.29** % RMS
Filter delay: gyro / D-term	**0.86 / 3.78** ms
Overshoot: roll / pitch / yaw	**13.3 / 20 / 1.5** %
Rise time: roll / pitch / yaw	**17.8 / 21.8 / 36.6** ms
Propwash: median wobble after a chop	**3.17** × calm (4 chops, 25% bad)
Battery wasted holding attitude	**2.07** %

Tune verdict: Roll Good Pitch OK Yaw Good

Top priorities

Roll: 20° phase margin (model range 17–20°) at the 17 Hz crossover, the filters take 31° of it
Even with no filtering at all this loop would only have 51°: the motors' spin-up lag and the ESC delay use most of the phase at 17 Hz, so every bit of filter delay shows.

Roll: lighter D-term filtering would buy 3° of phase margin
Raising the D-term low-pass cutoffs by 25% adds 3° for about +0.15% motor noise on this axis (predicted from your own gyro spectrum).

Pitch: 28° phase margin (model range 28–32°) at the 12 Hz crossover, the filters take 20° of it
Even with no filtering at all this loop would only have 48°: the motors' spin-up lag and the ESC delay use most of the phase at 12 Hz, so every bit of filter delay shows.

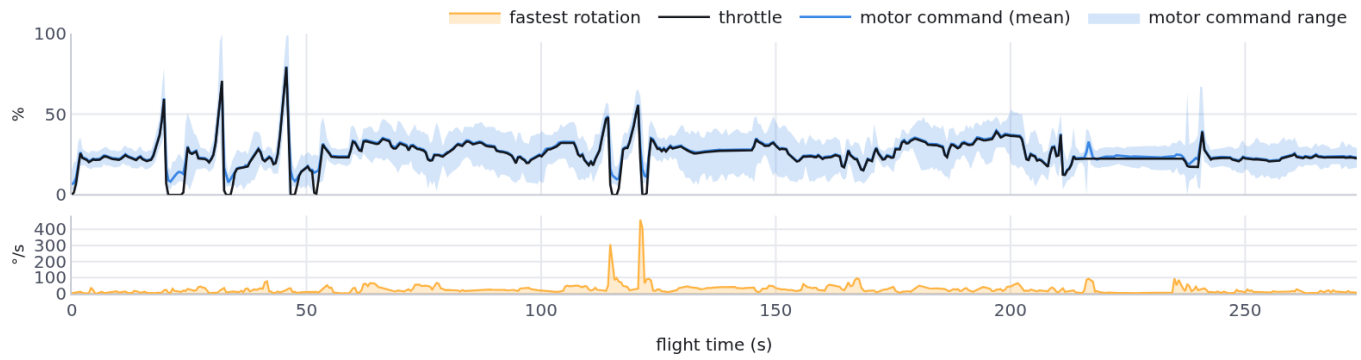
R2: frame vibration at 56 Hz (+8 dB, in 100% of the throttle range)
On roll, yaw · 0–30% throttle · the filters remove 1 dB

Pitch: 20% overshoot (90% range 16–29%)
It goes past the target before settling.

M3 (rear-left): 1.9× more once-per-turn vibration than the others (at least 1.8×)
Probably an out-of-balance prop or motor.

Propwash: 3.2× more wobble after throttle chops (4 chops)
Noticeable wobble when dropping into dirty air.

Flight at a glance

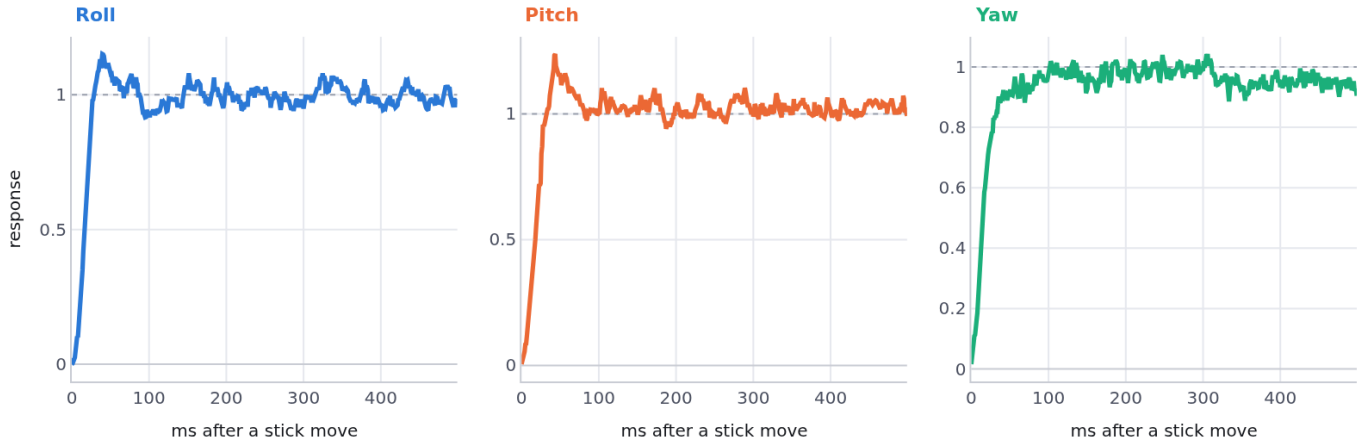


Top: throttle (black), average motor command (blue) and the spread between the busiest and the idlest motor (band). Bottom: the fastest rotation on any axis.

Stick response & the PID

How the quad follows your sticks, and what each PID term contributes

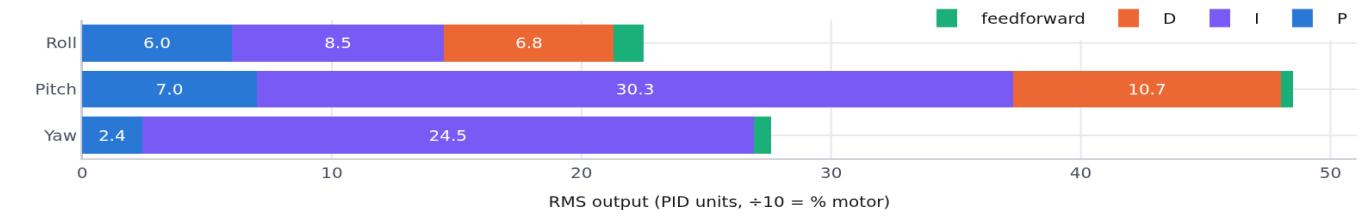
Step response



Median reaction to your stick moves (1.0 = exactly what you asked).

	RISE	OVERSHOOT	SETTLES	LATENCY	MOVES	VERDICT
Roll	17.8 ms	13.3 %	50 ms	17.8 ms	258	Good
Pitch	21.8 ms	20.0 %	58 ms	18.8 ms	127	OK
Yaw	36.6 ms	1.5 %	34 ms	15.8 ms	195	Good

What each PID term puts out



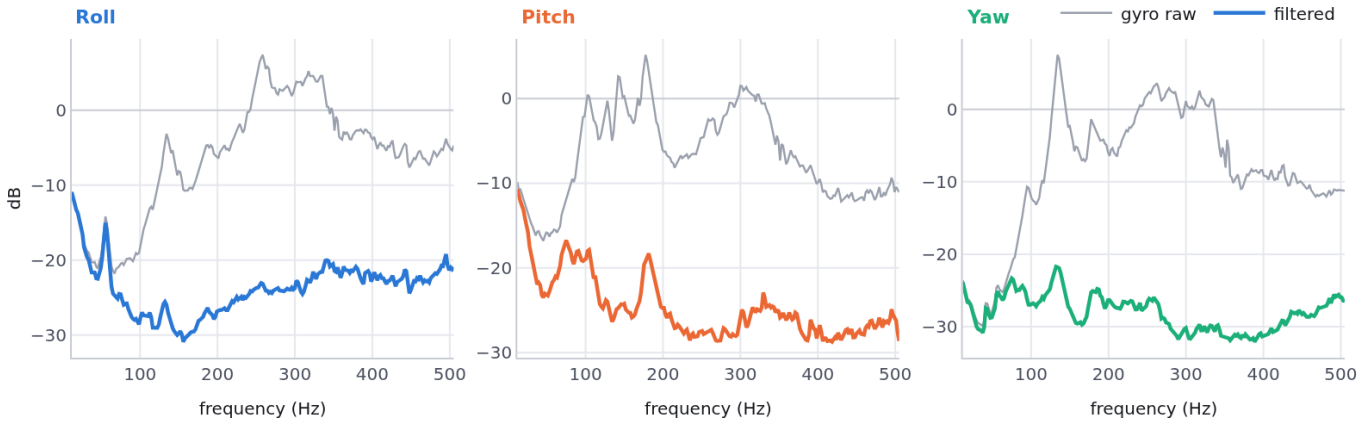
Size of each term's output over the flight. P and D fight disturbances, I holds trims, feedforward follows the sticks.

	D NOISE > 100 HZ	P NOISE > 100 HZ	OUTPUT AT THE LIMIT
Roll	0.5	0.6	0.0 %
Pitch	0.6	0.5	0.0 %
Yaw	–	0.3	0.0 %

Noise & filters

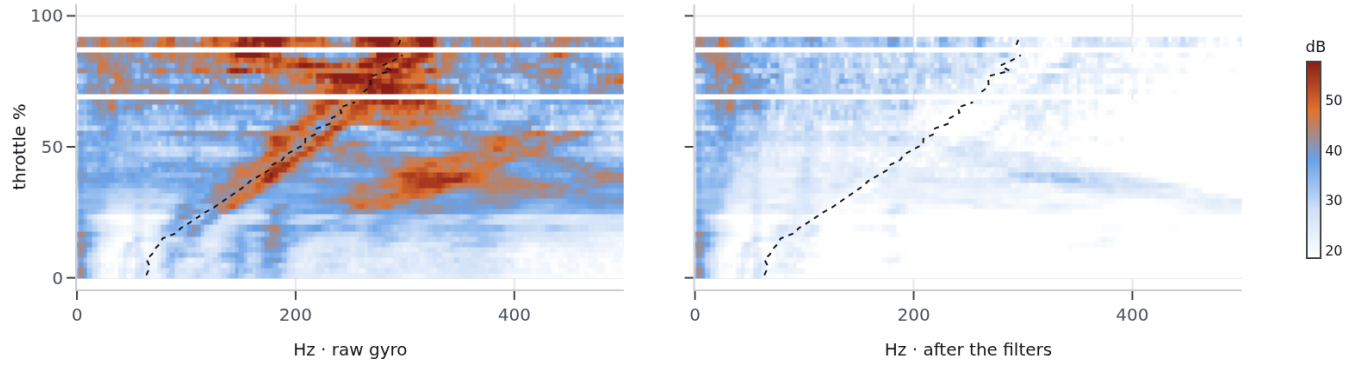
Where the vibration is, what the filters leave, and what their delay costs the PID

Gyro noise



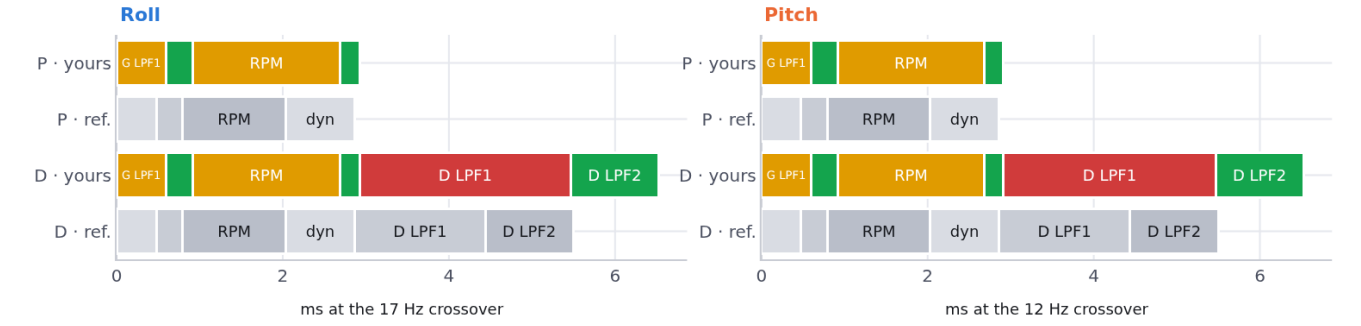
Raw gyro (grey) and after the filters (colour). Peaks that survive the filters reach the D term and the motors.

Noise by throttle (roll + pitch)



Each row is a throttle band; the dotted line is the motors' own frequency. Lines that follow it are motor noise (the RPM filter's job); fixed vertical lines are frame resonances.

Filter delay at the crossover · yours vs betaflight defaults



Each segment is one filter's delay where stability is decided. Green: like the reference or lighter; amber: up to 1.5× heavier; red: more than 1.5×. P path = what P and I see; D path adds the D-term filters.

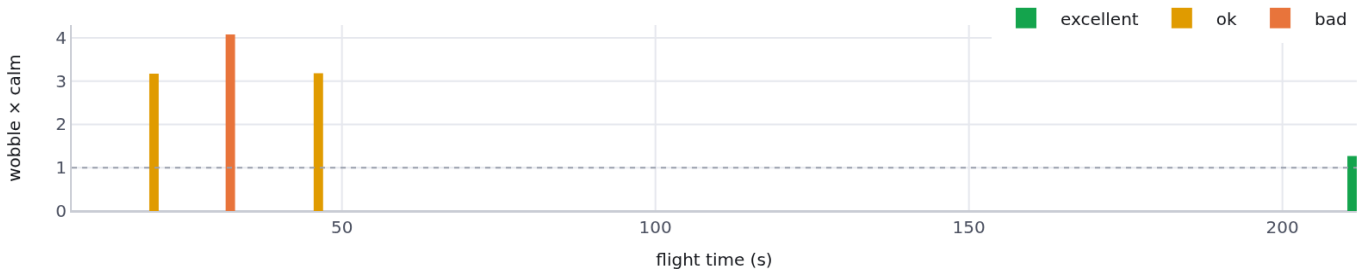
	CROSSOVER	PHASE MARGIN	MODEL RANGE	REFERENCE FILTERS	NO FILTERS	FILTERS TAKE
Roll	17.0 Hz	20°	17–20°	25°	51°	31°
Pitch	11.9 Hz	28°	28–32°	31°	48°	20°
Yaw	6.5 Hz	85°	84–85°	86°	96°	11°

Phase margin: how far the loop is from oscillating at the crossover (≥ 45° healthy, 30–45° rings a little, below 30° rings clearly). The model range is the margin over equally good fits with the motor lag at ±15% of its eRPM measurement.

Propwash & stability

The wobble after throttle chops, and whether anything oscillates on its own

Propwash: wobble after each throttle chop



× the calm level; dotted = calm. The wobble rings at about 23.7 Hz.

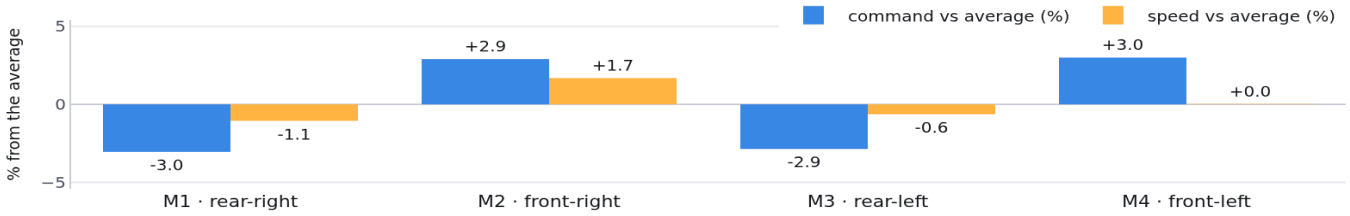
	TRACKING LAG	ERROR WHILE MOVING	ERROR WHEN CALM
Roll	19.3 ms	5.8 °/s	3.0 °/s
Pitch	20.9 ms	5.4 °/s	3.4 °/s
Yaw	12.3 ms	2.3 °/s	1.7 °/s

	STRONGEST 10–40 HZ WOBBLE, STICKS CENTRED	SIZE	STANDS OUT BY	
Roll	10.9 Hz	0.96 °/s	4.9 dB	none
Pitch	11.8 Hz	1 °/s	4.7 dB	none

Motors & power

Balance between the motors and where the battery goes

Motor balance



A motor that needs more command than the others, or spins faster, points to a heavier corner, a bent prop or a tired motor (positions as Betaflight numbers them).

HOVER COMMAND	THRUST / WEIGHT	EXTRA POWER: STEADY IMBALANCE	MANOEUVRES	CORRECTIONS	VIBRATION
23.8 %	8.6×	1.8 %	0.2 %	0.0 %	0.0 %

This quad's dynamics

Learned from the flight: what the PID simulator and the filter-delay analysis run on

	AUTHORITY	TYPICAL FOR THE SIZE	MOTOR LAG	SYSTEM DELAY	DAMPING	FIT	STICKS EXCITED UP TO
Roll	117	96	32 ms	2.0 ms	2.0	89 %	8 Hz
Pitch	86	96	42 ms	2.0 ms	0.0	86 %	8 Hz
Yaw	14	9	32 ms	2.0 ms	3.0	90 %	11 Hz

Motor lag measured from motor command vs eRPM: 37.9 ms at 116 Hz (6966 RPM), 49 ms at 105 Hz, 44 ms at 115 Hz, 36 ms at 144 Hz. Each axis may refine it by at most ±15%: stick inputs alone can't tell motor lag from authority.

Tune plan

In the order to change things. Ticked items are in the CLI block at the end.

⚠ Filters and PIDs both change: if the next flight feels worse, undo the PID step first (filter changes are the safer half).

1 Before the next flight: logging

So the next log can show what these changes did.

- **Weigh the quad once and enter the all-up weight**
  Hover power, battery waste in watts and the simulator's typical-quad comparison use it.
- **Log faster: this log runs at 1010 Hz, so nothing above 505 Hz is visible**
  For a filter flight, 2–4 kHz shows motor noise and its harmonics. In the Betaflight Blackbox tab set the rate to 1/2 or 1/1 (CLI: blackbox_sample_rate).

☐ **Turn on high-resolution logging**
Gyro and setpoint are logged in whole °/s; high resolution logs 0.1 °/s (check the name with: get blackbox_high_resolution).
Optional: check the setting name on your firmware first.
blackbox_high_resolution = 1

☐ **For a filter flight, log the dynamic-notch frequencies**
debug_mode = FFT_FREQ records where the dynamic notches sit, so the Spectrogram can show whether they follow the noise.
Optional: replaces any other debug mode you use.
debug_mode = FFT_FREQ

2 Mechanical first

Vibration and imbalance from the hardware make every filter and PID setting worse. Fix these before tuning around them.

- **M3 (rear-left): 1.9× more once-per-turn vibration than the others (at least 1.8×)**
  In this order: 1) fit a fresh prop; 2) check the prop nut is tight and the prop sits flat; 3) spin the motor by hand and look for a bent shaft or wobbling bell; 4) look for dirt or a loose magnet inside the bell. To tell prop from motor: swap the M3 prop with a quiet motor's prop of the same spin direction and log again. If the vibration follows the prop, it's the prop. If it stays at that corner, it's the motor or the arm (part of a difference can also come from how far that arm is from the flight controller).
- **R2: frame vibration at 56 Hz (+8 dB, in 100% of the throttle range)**
  It sits close to the range the controller works in, so a notch here costs response: a mechanical fix is better. It's outside the dynamic notch range (70–500 Hz): to let it track this, set dyn_notch_min_hz ≤ 48.

3 Filters

Filters decide how much noise and delay the PID loop sees, so they come before the PIDs.

☐ **Static gyro notch on R2 at 56 Hz (Q 5.7)**
R2 at 56 Hz stands +8 dB on roll, yaw, the filters remove only 1 dB there; measured peak width ≈ 8 Hz → Q ≈ 5.7
This is close to the frequencies the controller works with: a notch here costs noticeable response. Fix it mechanically if you can (see the finding).
gyro_notch1_hz = 56 gyro_notch1_cutoff = 52

☑ **Let the RPM filter work fully where this quad flies**
Its fade starts at 75 Hz + 50 Hz while the motors hover near 114 Hz, so the notches run at only 79% strength.
rpm_filter_min_hz = 40

☐ **Noise reaching the motors: 1.3% RMS above 80 Hz, mostly from D**
Filter the D-term more (set dterm_lpf1_dyn_min_hz = 51, dterm_lpf1_dyn_max_hz = 102; set dterm_lpf2_static_hz = 128: D-term delay 3.78 → ~4.26 ms) or lower D a little.
Optional: small gain, a little more delay.
dterm_lpf1_dyn_min_hz = 51 dterm_lpf1_dyn_max_hz = 102 dterm_lpf2_static_hz = 128

☐ **Roll: lighter D-term filtering would buy 3° of phase margin**
Raising the D-term low-pass cutoffs by 25% adds 3° for about +0.15% motor noise on this axis (predicted from your own gyro spectrum).
Optional: from the Filter delay tab's loop model; try it on its own flight.
dterm_lpf1_dyn_min_hz = 75 dterm_lpf1_dyn_max_hz = 150 dterm_lpf2_static_hz = 188

4 PIDs

One careful step, sized for your props. Fly, log, check again.

- **Roll: keep the PIDs**
  already good: keep it

☑ **Pitch: P 61 → 56**
20% overshoot but D-term is already noisy → less P (−8%)
p_pitch = 56

- **Yaw: keep the PIDs**
  already good: keep it

6 Worth trying later

Smaller wins, or ideas to test one at a time once the above is done.

- **Propwash: 3.2× more wobble after throttle chops (4 chops)**
  In order: During these chops a motor sat at its minimum 6% of the time, so the mixer couldn't slow it to correct: raise dyn_idle_min_rpm (now 25) and/or thrust_linear (now 25); More D max (+10–15%) raises damping on fast changes like these; less filter delay helps too (Noise tab).
  One option among several: try it on its own flight after the steps above.
- **Pitch: I-term constantly holds -27 (the front needs a steady push)**
  Move weight away from the front until the Motors tab shows the load evenly shared. If it only appears in fast cruise, it's aerodynamic and harmless.
- **Roll: the strongest ripple in the error (54 Hz) is vibration, not the tune**
  See the Noise tab's filter planner for a notch or filter change.
- **Pitch: D is mostly vibrating (54% of it above 80 Hz, strongest at 88 Hz)**
  See 'Noise reaching the motors' and the filter planner in the Noise tab.
- **Front motors work 5.9% harder than the rear pair**
  Move the battery (or other heavy parts) away from the front until the difference is under ~1%.
- **Unequal motor speeds cost about 2.1% extra prop power**
  The steady part is a CG offset, a bent or mismatched prop, or a weaker motor: rebalance the battery position and swap props to find it.

CLI for the ticked items

```
# BBX Analyzer v2.4 tune plan · Fawkes · 2026-10-13
# 3. Filters
set rpm_filter_min_hz = 40
# 4. PIDs
set p_pitch = 56
save
```

Next test flight

1. Hover 20–30 s, land, touch each motor: warm is fine, too hot to hold means back off D or filtering.
2. Fly 5 sharp flicks per axis (roll, pitch, yaw) with the other sticks still: the Step response tab needs them.
3. Do a slow throttle sweep from hover to full and back: the Spectrogram shows what the filters do across the range.
4. Do 4–6 throttle chops from 60–80% to zero with the sticks still: the Propwash tab scores them.
5. After the mechanical fix, check Motor health again: the per-motor vibration should even out.
6. Load the new log here and use Compare with the previous one.

All findings

Grouped by analysis; the app has the charts and the full explanations.

Stick response

Good **Roll: no problems in the step response (13% overshoot, 18 ms rise, settles at 0.99)**

Fast and well damped for this selection.

Check **Pitch: 20% overshoot (90% range 16–29%)**

It goes past the target before settling.

Try: Options: D +10–15% (D 44 → 49), or P −10% (P 61 → 55), or FF −15% (125 → 106) if it's mostly on quick flicks.

Good **Yaw: no problems in the step response (2% overshoot, 37 ms rise, settles at 0.96)**

Fast and well damped for this selection.

PID terms

Info **Pitch: I-term constantly holds -27 (the front needs a steady push)**

The quad needs a steady correction on this axis just to fly straight.

Try: Move weight away from the front until the Motors tab shows the load evenly shared. If it only appears in fast cruise, it's aerodynamic and harmless.

Info **Roll: the strongest ripple in the error (54 Hz) is vibration, not the tune**

It sits on a frame resonance; P passes it straight to the motors.

Try: See the Noise tab's filter planner for a notch or filter change.

Info **Pitch: D is mostly vibrating (54% of it above 80 Hz, strongest at 88 Hz)**

The strongest fast part of D sits at 88 Hz: no motor order or resonance.

Try: See 'Noise reaching the motors' and the filter planner in the Noise tab.

Info **Flown 32% in a self-levelling mode (angle 32%, horizon 0%)**

There the roll/pitch setpoint comes from the levelling controller, not straight from the sticks.

Info **How hard each term works (RMS in flight)**

Roll: P 6.0 I 8.5 D 6.8 FF 1.2 · Pitch: P 7.0 I 30.3 D 10.7 FF 0.5 · Yaw: P 2.4 I 24.5 D 0.0 FF 0.7

Noise & filters

Good **Filter delay: gyro 0.86 ms, D-term 3.78 ms**

Less delay = sharper and better in propwash; more delay = smoother and quieter.

Try: Guide for 7″ props (bigger props react slower, so they tolerate more delay): gyro under 1.4 ms is light, over 2.8 ms heavy. D-term under 3.5 ms light, over 5.6 ms heavy.

Info **Noise reaching the motors: 1.3% RMS above 80 Hz, mostly from D**

Comes from D (it amplifies whatever fast gyro noise is left).

Try: Filter the D-term more (set dterm_lpf1_dyn_min_hz = 51, dterm_lpf1_dyn_max_hz = 102; set dterm_lpf2_static_hz = 128: D-term delay 3.78 → ~4.26 ms) or lower D a little.

Good **The RPM filter removes motor noise well**

×1 (≈121 Hz): −20 dB, ×2 (≈242 Hz): −26 dB, ×3 (≈357 Hz): −25 dB

Info **R1: 135 Hz frame vibration, already filtered out (−22 dB)**

On roll, pitch, yaw · 5–50% throttle · +11 dB in the raw gyro.

Try: Nothing to do for the tune. If it grows on later flights, look for whatever loosened.

Check **R2: frame vibration at 56 Hz (+8 dB, in 100% of the throttle range)**

On roll, yaw · 0–30% throttle · the filters remove 1 dB

Try: It sits close to the range the controller works in, so a notch here costs response: a mechanical fix is better. It's outside the dynamic notch range (70–500 Hz): to let it track this, set dyn_notch_min_hz ≤ 48.

Info **R3: 180 Hz frame vibration, already filtered out (−23 dB)**

On pitch, yaw · 5–45% throttle · +8 dB in the raw gyro.

Try: Nothing to do for the tune. If it grows on later flights, look for whatever loosened.

Info **This log can't show anything above 505 Hz**

Gyro lpf2 and dyn notch max sit above what a 1010 Hz log can show.

Try: For a filter-tuning flight, set blackbox_sample_rate to 1/2 or 1/1 (2–4 kHz).

Filter delay & stability

Check **Roll: 20° phase margin (model range 17–20°) at the 17 Hz crossover, the filters take 31° of it**

Even with no filtering at all this loop would only have 51°: the motors' spin-up lag and the ESC delay use most of the phase at 17 Hz, so every bit of filter delay shows.

Check **Roll: lighter D-term filtering would buy 3° of phase margin**

Raising the D-term low-pass cutoffs by 25% adds 3° for about +0.15% motor noise on this axis (predicted from your own gyro spectrum).

Try: set dterm_lpf1_dyn_min_hz = 75, dterm_lpf1_dyn_max_hz = 150; set dterm_lpf2_static_hz = 188. Hover, punch a few times, land and feel the motors.

Info **Roll: with these filters no P/D setting reaches 45°**

Scaling P and D together never gets above 33° with the current filters; without filtering ×2.50 would still keep 45°.

Info **Roll: the RPM filter (3 harmonics × 4 motors) costs 1.78 ms at the crossover (27% of the filter delay)**

Notches delay low frequencies too: each one adds about 1 / (Q · 2π · its frequency) of delay well below its centre.

Check **Pitch: 28° phase margin (model range 28–32°) at the 12 Hz crossover, the filters take 20° of it**

Even with no filtering at all this loop would only have 48°: the motors' spin-up lag and the ESC delay use most of the phase at 12 Hz, so every bit of filter delay shows.

Info **Pitch: with these filters no P/D setting reaches 45°**

Scaling P and D together never gets above 32° with the current filters; without filtering ×2.50 would still keep 45°.

Info **Pitch: the RPM filter (3 harmonics × 4 motors) costs 1.77 ms at the crossover (27% of the filter delay)**

Notches delay low frequencies too: each one adds about 1 / (Q · 2π · its frequency) of delay well below its centre.

Good **Yaw: 85° phase margin (model range 84–85°) at the 6 Hz crossover, the filters take 11° of it**

Without filtering it would have 96°; the filters use 22% of the phase the loop can spare above the 45° target.

Good **Yaw: filter delay is not what limits this PID**

It keeps 85° of phase margin; lighter filtering would change little.

Info **Cross-check: propwash rings at 24 Hz, the model's resonance sits at 20, 15 Hz (roll, pitch)**

A loop with little phase margin rings near its sensitivity peak, and propwash is what excites it.

Good **Model check: predicted PID noise 0.88% vs logged 1.18%**

How closely the filter + PID model reproduces the high-frequency part of the logged PID output (above 80 Hz, as motor %).

Propwash

Check **Propwash: 3.2× more wobble after throttle chops (4 chops)**

Noticeable wobble when dropping into dirty air.

Try: In order: During these chops a motor sat at its minimum 6% of the time, so the mixer couldn't slow it to correct: raise dyn_idle_min_rpm (now 25) and/or thrust_linear (now 25); More D max (+10–15%) raises damping on fast changes like these; less filter delay helps too (Noise tab).

Info **2 chop(s) left out: the sticks were moving**

Only wobble with centred, still sticks counts as propwash.

Info **Stick-to-motion delay: roll 19.3 ms, pitch 20.9 ms, yaw 12.3 ms**

How long the quad takes to follow your sticks.

Info **Tracking error during stick moves: roll 5.8, pitch 5.4, yaw 2.3 °/s RMS**

Average gap between what you asked and what the quad did.

Motors & power

Check **M3 (rear-left): 1.9× more once-per-turn vibration than the others (at least 1.8×)**

Probably an out-of-balance prop or motor.

Try: In this order: 1) fit a fresh prop; 2) check the prop nut is tight and the prop sits flat; 3) spin the motor by hand and look for a bent shaft or wobbling bell; 4) look for dirt or a loose magnet inside the bell. To tell prop from motor: swap the M3 prop with a quiet motor's prop of the same spin direction and log again. If the vibration follows the prop, it's the prop. If it stays at that corner, it's the motor or the arm (part of a difference can also come from how far that arm is from the flight controller).

Info **Front motors work 5.9% harder than the rear pair**

Weight toward the front, or a steady nose-down moment from the frame and camera in forward flight.

Try: Move the battery (or other heavy parts) away from the front until the difference is under ~1%.

Info **Holding attitude costs about 2.4% extra battery power (range 2.1–2.7%)**

Some energy goes into driving the motors up and down without making thrust.

Info **Unequal motor speeds cost about 2.1% extra prop power**

For the same total thrust, motors spinning at different speeds draw more power than motors at the same speed.

Try: The steady part is a CG offset, a bent or mismatched prop, or a weaker motor: rebalance the battery position and swap props to find it.

Info **Thrust headroom about 8.6:1 (extrapolated)**

Full-throttle speed 321 Hz vs hover 109 Hz (hover command 24%). Thrust goes with speed², so T/W ≈ (full / hover)².

Good **All motors spin equally fast for the same command**
M1 -1.1% · M2 +1.7% · M3 -0.6% · M4 +0.0%

Good **No desyncs or speed dropouts**
Every motor followed its command.